

Thomas GARABEDIAN

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EDUCATION

- Institut Polytechnique de Paris, École Polytechnique** Sep. 2026
Master 2 Data Science (M2DS) Palaiseau, France
- Targeted coursework: Machine Learning, Deep Learning, GNNs, LLMs, Computer Vision.
- Institut Polytechnique de Paris, Télécom SudParis** 2024 – 2027
Engineering Degree in Digital Technologies Évry, France
- GPA: 3.7/4.0**, top 15% among 220 students.
 - Relevant coursework: Applied Statistics, Optimization, Signal Processing, Data Mining, Cryptography, Secure Communications, Data Analysis, Stochastic Processes.
- CPGE Lycée Thiers & Prytanée National Militaire** 2021 – 2024
Intensive undergraduate program preparing for French engineering school, PCSI then PC track France

EXPERIENCE

- CIEDS IP Paris Project | Drone Swarm Automation** Jan. – June 2026
ENSTA Paris Palaiseau, France
- Designed a computer vision pipeline to detect vehicles and buildings in aerial imagery, in a multi-drone surveillance context. Presented the results to CIEDS IP Paris.
 - Contributed to the swarm coordination logic: target assignment, trajectory planning and scenario simulation. Compared object detection models: CNN, YOLO, DETR.
- Engineering Internship | ALSEAMAR, Alcen Group** June – Aug. 2025
Embedded systems, RF and industrial software Six-Fours, France
- Developed control software for a high-frequency power amplifier used in an industrial environment and for submarine antenna testing.
 - Implemented equipment communication, supervision, acquisition, data logging and diagnostic modules.
 - Carried out validation tests to improve system control reliability and measurement traceability.
 - Worked at the interface of Python, embedded systems, RF electronics, signal processing and industrial constraints.
- Special Military Preparation** 2023
École du Génie Thorée-les-Pins, France
- Intensive training in leadership, decision-making under constraints and defense fundamentals.

AI & DATA SCIENCE PROJECTS

- HAKS Hackathon, Airbus x IBM | 1st Prize, €2,500** June 2026
Télécom Paris Palaiseau, France
- Developed a corrosion prediction model for an aeronautical use case, combining FT-Transformer and XGBoost.
 - Built an end-to-end machine learning pipeline, from data preprocessing and feature engineering to model comparison, calibration and reliability-focused evaluation.
 - Optimized and probabilistically evaluated the model with a Brier Score of 0.150.
- D4GEN Hackathon, Genopole x AWS | 1st Prize, €10,000** May – June 2026
Genopole d'Évry Évry, France
- Designed a human-in-the-loop tool to accelerate expert annotation for animal species monitoring.
 - Built a pipeline combining unsupervised clustering, clip export, spectrograms and an annotation API.
 - Fine-tuned SAM 3 for segmentation and visual identification support in a weakly labeled data context.

SKILLS

- AI & Computer Vision:** object detection, segmentation, classification, CNN, YOLO, DETR, SAM 3, Transformers, FT-Transformer, XGBoost, LightGBM, CatBoost.
- Model Evaluation:** IoU, precision, recall, error analysis, probabilistic calibration, Brier Score, WAPE.
- Languages:** French (Native), English (C1), Spanish (C1).
- Sports:** Competitive tennis, official referee, ranked first in training cohort.